

FLOATING POINT REPRESENTATION

PRACTICE SHEET • MEDIUM LEVEL

ResourceOS

1. What is the hexadecimal representation of the IEEE 754 single-precision float for decimal number -12.375?

- (A) C1460000
- (B) 41460000
- (C) C1450000
- (D) C2460000

2. A 16-bit custom floating-point format uses 1 sign bit, 5 exponent bits (bias = 15), and 10 mantissa bits. What is the value of 0 10010 1000000000?

- (A) 12.0
- (B) 6.0
- (C) 1.5
- (D) 24.0

3. What is the smallest positive normalized number in IEEE 754 single-precision format?

- (A) 1.0×2^{-126}
- (B) 1.0×2^{-127}
- (C) 1.0×2^{-128}
- (D) 2^{-23}

4. What is the smallest positive subnormal number in IEEE 754 single-precision format?

- (A) 2^{-149}
- (B) 2^{-126}
- (C) 2^{-127}
- (D) 2^{-150}

5. What is the Machine Epsilon (ϵ) for IEEE 754 single-precision representation?

- (A) 2^{-23}
- (B) 2^{-24}
- (C) 2^{-52}
- (D) 2^{-127}

6. If an 8-bit floating-point format has 1 sign bit, 4 exponent bits (bias = 7), and 3 mantissa bits, what decimal value does bit pattern 0 0000 011 represent?

- (A) 0.0234375 (3/128)
- (B) 0.046875 (3/64)
- (C) 0.09375 (3/32)
- (D) 0.375

7. What is the result of adding 1.0×2^{20} and 1.0×2^0 in IEEE 754 single-precision format?

- (A) $1.0 \times 2^{20} + 1$
- (B) Exact sum (1048577.0)
- (C) 1.0×2^{20} (due to alignment loss)
- (D) Overflow exception

8. Which of the following bit patterns represents a Signaling NaN (SNaN) in single-precision IEEE 754?

- (A) Exp = 255, Mantissa = 01000...
- (B) Exp = 255, Mantissa = 10000...
- (C) Exp = 255, Mantissa = 00000...
- (D) Exp = 0, Mantissa = 01000...

9. Convert decimal fraction 0.1 to binary. How is it characterized in binary floating point?

- (A) Infinitely repeating fraction: $0.000110011..._2$
- (B) Terminating binary fraction: 0.00011_2
- (C) Exact binary fraction: 0.001_2
- (D) Denormalized number

10. In IEEE 754 default rounding mode (Round to Nearest, Ties to Even), what is the 3-bit round result of binary 1.0111_2 truncated to 2 fractional bits?

- (A) 1.10_2 (B) 1.01_2
(C) 1.11_2 (D) 2.00_2

11. What is the stored exponent for the largest finite normalized number in IEEE 754 single precision?

- (A) 254 (FE_{16}) (B) 255 (FF_{16})
(C) 127 ($7F_{16}$) (D) 128 (80_{16})

12. What decimal value is represented by the IEEE 754 single-precision hex value $0x3F800000$?

- (A) 1.0 (B) 0.5
(C) 2.0 (D) 1.5

13. What decimal value is represented by the IEEE 754 single-precision hex value $0xBF800000$?

- (A) -1.0 (B) -0.5
(C) -2.0 (D) -1.5

14. In a floating-point system, what phenomenon occurs when subtracting two almost identical numbers?

- (A) Catastrophic Cancellation (B) Overflow
(C) Gradual Underflow (D) Exponent Wrap

15. What is the bias for a IEEE 754-style float with a 6-bit exponent field?

- (A) 31 (B) 32
(C) 63 (D) 15

16. What is the purpose of using a biased exponent instead of 2's complement for the exponent field?

- (A) Allows fast comparison using standard integer magnitude comparisons (B) Eliminates the need for a sign bit in the fraction
(C) Increases the maximum achievable precision (D) Simplifies mantissa multiplication

17. Which statement regarding subnormal (denormalized) numbers is CORRECT?

- (A) They fill the gap between 0 and the smallest normalized number (B) They have an implicit leading 1 bit before the binary point
(C) Their stored exponent field value is 1 (D) They represent numbers greater than 1.0

18. Express decimal 0.0078125 ($1/128$) as an IEEE 754 single-precision binary scientific number.

- (A) $1.0_2 \times 2^{-7}$ (B) $1.0_2 \times 2^{-8}$
(C) $1.0_2 \times 2^{-6}$ (D) $0.1_2 \times 2^{-7}$

19. What is the stored 8-bit binary exponent for decimal 0.0078125 (1/128) in single precision?

- (A) 01111000_2 (120) (B) 01110111_2 (119)
(C) 01111001_2 (121) (D) 10000111_2 (135)

20. Which floating-point operation can produce a NaN result according to IEEE 754?

- (A) $0.0 / 0.0$ (B) $1.0 / 0.0$
(C) $1.0 / \infty$ (D) $\infty + \infty$

21. How many total distinct bit representations exist for zero (+0 and -0) in IEEE 754 standard?

- (A) Exactly 2 (B) Exactly 1
(C) 4 (D) Depends on precision

22. What is the result of IEEE 754 operation $+\infty + (-\infty)$?

- (A) NaN (B) 0.0
(C) $+\infty$ (D) Overflow exception

23. In IEEE 754 double-precision format, what is the bias value for the exponent?

- (A) 1023 (B) 127
(C) 2047 (D) 511

24. Convert decimal 100.25 to IEEE 754 single-precision binary normalized mantissa.

- (A) 100100010000000000000000 (B) 100100100000000000000000
(C) 001001000000000000000000 (D) 110010010000000000000000

25. What is the approximate range of values printable in IEEE 754 single precision?

- (A) 1.4×10^{-45} to 3.4×10^{38} (B) 4.9×10^{-324} to 1.8×10^{308}
(C) 1.0×10^{-127} to 1.0×10^{128} (D) 2.0×10^{-23} to 2.0×10^{23}

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(C)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)